1. Discrete Random Variables

1.1. Probability Distribution Tables

- The probability distribution of a discrete random variable lists the values that a random variable can take and shows the corresponding probabilities.
- The sum of all the probabilities in a probability distribution table always adds upto 1.
  - This is useful when finding the values of unknown probabilities in the table.

Let X represent a discrete random variable.

X	1	2	3	4
P(X)	$\frac{2}{40}$	$\frac{6}{40}$	$\frac{12}{40}$	$\frac{20}{40}$

- In this example, $P(X=1)$ would represent the probability when the random variable X takes a value of 1. Looking at the probability distribution table, $P(X=1)=\frac{2}{40}$.

Mean and Variance

- The mean of a random variable is also called its expectation, or expected value.
  - $E(X)$ represents the expected value of the random variable X.

To find $E(X)$:

- Multiply the values that the random variable X can take with the corresponding probability, sum up all the results to get the expectation.

$$\mu = E(x) = \sum x_i p_i \quad \text{[Given in MF19]}$$

- The variance of a random variable X is denoted by Var(X).

To find $Var(X)$:

- Find the sum of the product of the squared values of the values the random variable X can take with the corresponding probability. Subtract this value with the mean squared.

$$\sigma^2 = \sum x_i^2 p_i - (E(X))^2 \quad \text{[Given in MF19]}$$

1.2. Examples

Example 1

A fair 6-sided dice has the numbers 1, 2, 2, 3, 3, 3 on its 6 sides. The die is rolled twice. The random variable X denotes the sum of of the two numbers obtained. Draw the probability distribution table for X, and find the mean and variance of X.

Answer

The random variable X is the sum of two numbers obtained after rolling the die. Using this, we know that the values X can take are:

$$1+1=2,\ 1+2=3,\ 1+3=4,\ 2+3=5,\ 3+3=6$$

- Note that other combinations exist to obtain the same values listed above.

Now we can find the probability of the random variable X taking up each value using a probability space diagram. Each row and column, apart from the first one, represents the sum of the two faces obtained.

	1	2	2	3	3	3
1	2	3	3	4	4	4
2	3	4	4	5	5	5
2	3	4	4	5	5	5
3	4	5	5	6	6	6
3	4	5	5	6	6	6
3	4	5	5	6	6	6

Using

$$\text{Probability} = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

We can draw the probability distribution table for X as:

x	2	3	4	5	6
p	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$

- We also note that all the probabilities in the table add upto 1.

To find the mean and variance, we use the probability distribution table.

$$E(X) = \frac{2\times 1}{36} + \frac{3\times 4}{36} + \frac{4\times 10}{36} + \frac{5\times 12}{36} + \frac{6\times 9}{36} = \frac{14}{3}$$

To find the variance, we first find $\sum x_i^2 p_i$.

$$\sum x_i^2 p_i = \frac{(2)^2\times 1}{36} + \frac{(3)^2\times 4}{36} + \frac{(4)^2\times 10}{36} + \frac{(5)^2\times 12}{36} + \frac{(6)^2\times 9}{36} = \frac{206}{9}$$

We subtract this value from the mean squared to find the variance

$$Var(X) = \frac{206}{9} - \left(\frac{14}{3}\right)^2 = \frac{10}{9}$$

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Example 2

The random variable X takes the values -1, 1, 2, 3 only. The probability that X takes the value x is

$$P(X = x) = kx^2 \quad [\text{Where } k \text{ is a constant}]$$

Draw the probability distribution table for X, in terms of k, and find the value of k.

Answer

We can draw the probability distribution table by inputting the values which X can take.

$$P(X = -1) = k(-1)^2 = k,\ P(X = 1) = k(1)^2 = k$$

$$P(X = 2) = k(2)^2 = 4k,\ P(X = 3) = k(3)^2 = 9k$$

So the probability distribution table is:

x	-1	1	2	3
p	k	k	$4k$	$9k$

To find the value of k, we make use of the fact that the sum of probabilities in a probability distribution table must add upto 1.

$$k + k + 4k + 9k = 1 \implies 15k = 1$$

Solving for k:

$$k = \frac{1}{15}$$

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